Flowers recognition
Workshop 2 - Part 2

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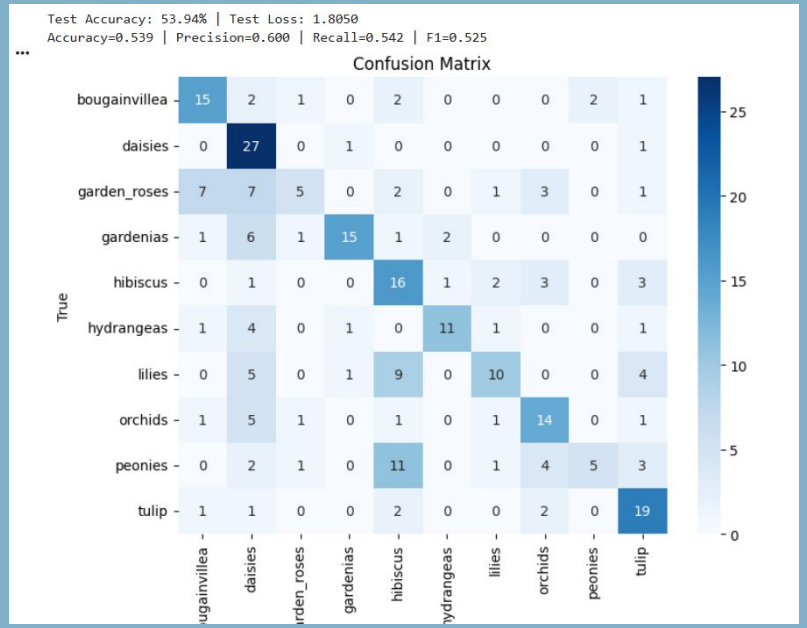
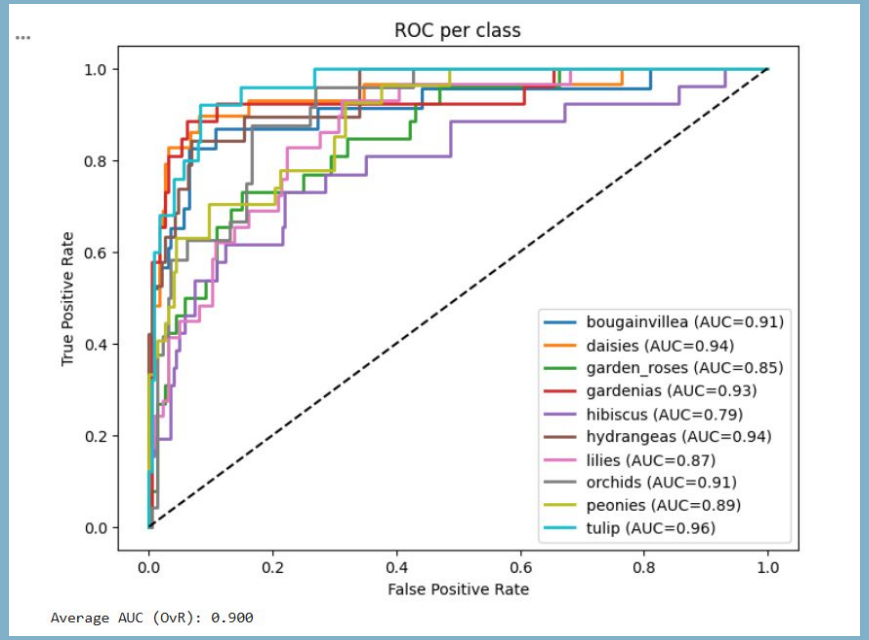
Our model before shortcuts





The model showed signs of learning but remained limited in overall performance, with an accuracy around 54%. ROC curves and AUC values suggested decent class separability but with considered fluctuations. The confusion matrix showed good alignment comparing predicted class probabilities and true labels but there were some misclassifications and uneven prediction reliability.

Our model with shortcuts

Shortcuts

Then we added shortcut connections between convolutional layers of the model. We did this by:

- Implementing shortcut connections inline, where the output of the first convolution in a block is stored and later added to the output of the second convolution in the same block.
- Each block consists of two convolutional layers with batch normalization and SiLU activation.
- After adding the shortcut to the main path, we applied another SiLU activation, followed by max pooling and dropout to reduce overfitting.
- We arranged three of these blocks with increasing filter sizes (64, 128 and 256) to progressively increase the abstraction and complexity of the model.

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Test Accuracy: 62.80% | Test Loss: 1.7227
Accuracy = 0.628 | Precision = 0.665 | Recall = 0.629 | F1 = 0.628
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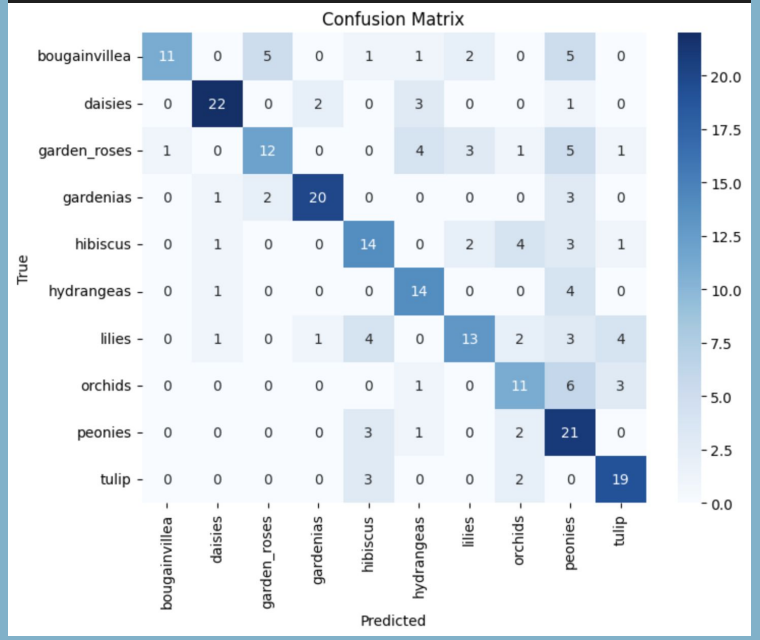

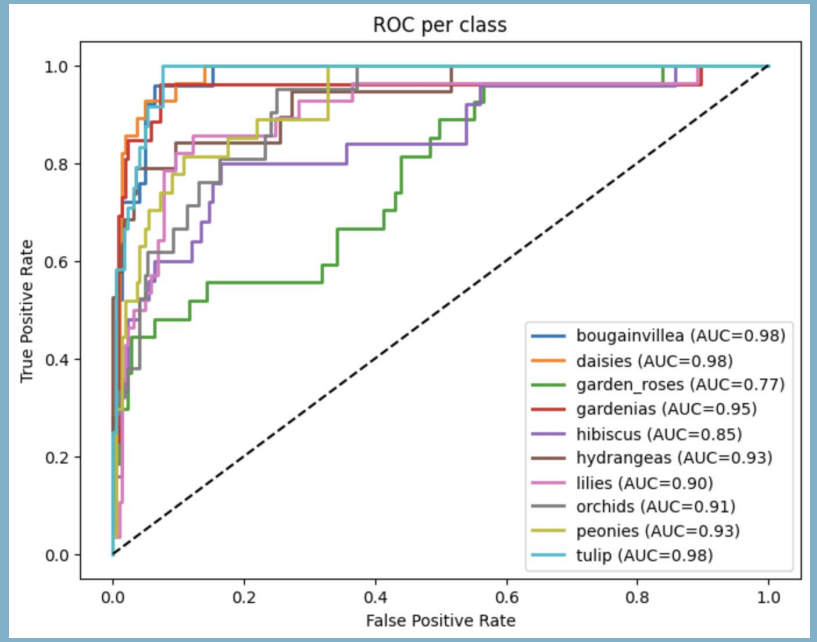

We can see great improvements. Accuracy raised up to 62.8%, as well as a F1 score of 0.628 and achieving almost perfect AUCs (e.g., tulip, daisies). As observed, ROC curves are consistent and with less fluctuations, stabilizing at early stages. The confusion matrix revealed more concentrated correct predictions and fewer misclassifications.

The End

Thank you for your time